Manish Rama Gopal Nadella

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SUMMARY

Machine Learning Engineer with 2.5 years of experience designing, building, and deploying end-to-end machine learning pipelines and predictive models. Expertise in statistical analysis, experimentation, and automation, with hands-on work in computer vision, NLP, LLM fine-tuning, and time-series domains. Proven track record improving anomaly detection outcomes by 20–25% and reducing failure diagnosis time through large-scale data analysis and hypothesis-driven experimentation.

EDUCATION

University of Michigan, Dearborn 09/2024 – 04/2026
Masters of Science in Artificial Intelligence GPA: 4.0/4.0
Courses: *Artificial Intelligence, Algorithm Design and Analysis, Software Engineering, Text Mining & Infor. Retrieval, Intelligent Systems, Robot Vision.*

Amrita Vishwa Vidyapeetham University, Coimbatore, India 07/2020 - 07/2024
B. Tech in Computer Science Engineering (Artificial Intelligence) GPA: 8.30/10
Courses: *Machine Learning, Deep Learning, Natural Language Processing, Signal and Image Processing, Computer networks, Reinforcement Learning, Robotic Systems and Drones, Mathematics for Intelligent Systems.*

PROFESSIONAL EXPERIENCE

Machine Learning / Data Analyst 05/2025 – 10/2025
Freudenberg Xalt Energy Systems – Michigan, USA

- Designed, deployed, and maintained end-to-end MLlearning pipelines in **Python** and **scikit-learn** to analyze battery performance data, generating actionable predictive insights that improved anomaly detection and performance monitoring by **~20–25%.**
- **Analyzed large-scale time-series battery data** to identify performance degradation patterns and detect potential system issues early, reducing failure diagnosis time by **~15%** and improving overall system reliability.

Machine Learning Research Engineer (LLMs) 12/2024 – 08/2025
University of Michigan – Michigan, USA

- **Studied and evaluated different Large Language Model (LLM) architectures** for their ability to understand and reason over human-written code, conducting quantitative comparisons using statistical metrics to assess code comprehension, logical flow, and error identification.
- **Experimented with fine-tuning LLM architectures on code-related data** using **PyTorch** and **TensorFlow**, improving model adaptation efficiency through structured experiments and evaluation workflows.

Deep Learning Engineer 09/2023 – 06/2024
Aptagrim Pvt. Ltd. – Hyderabad, India

- Built and deployed deep learning object detection and computer vision models in Python with PyTorch, improving accuracy by 15–20% with near real-time inference.
- Designed end-to-end ML pipelines including data ingestion with pandas, preprocessing, feature engineering, training with TensorFlow, and evaluation for image and text datasets to support scalable predictive analytics.
- Developed scalable data workflows using Pandas, MongoDB, and Snowflake to support analytics and ML experimentation on large structured and unstructured data.

PROJECTS

- Floodlight: Semi-supervised ResNet-based model for flood imagery classification; integrated VQA and LLM APIs for image-aware question answering. This work is also published at a notable conference (https://doi.org/10.1007/978-981-96-0451-7_11)
- Satellite Image Dehazing: Benchmarked ML and image processing dehazing techniques; build synthetic hazy/ground-truth data pipeline for evaluation.

SKILLS

Programming Languages: Python, C/C++, SQL | Machine Learning and AI: Machine Learning, Deep Learning, Computer Vision, NLP, LLMs | Frameworks: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy | Data management: MongoDB, Snowflake, Statistic